

$$b) A = \{w \in \{0,1\}^* / |w| \text{ is odd}\}$$

$$S \rightarrow 0S0 \mid 0S1 \mid 1S0 \mid 1S1 \mid 0 \mid 1$$

if we want the even length
we remove $(0 \mid 1)$ and add ϵ

$$B = \{w \in \{0,1\}^* / w = w^R \text{ and } |w| \text{ is even}\} \quad w^R = \text{reverse of } w \text{ (palindrome)}$$

$$S \rightarrow 1S1 \mid 0S0 \mid \epsilon$$

$\hookrightarrow 0 \mid 1$ (if $|w|$ odd)

$\hookrightarrow \epsilon \mid 0 \mid 1$ (if it doesn't matter even or odd)

$$C = \{a^i b^j c^k / i, j, k \geq 0 \text{ and } i=j^0 \text{ or } i=k\}$$

$$C_1 = \{a^i b^j c^k / i, j, k \geq 0 \text{ and } i=j\}$$

$$C_2 = \{a^i b^j c^k / i, j, k \geq 0 \text{ and } i=k\}$$

$$S_1 \rightarrow xy$$

$$x \rightarrow axb \mid \epsilon$$

$$y \rightarrow cy \mid \epsilon$$

$$S_2 \rightarrow aS_2c \mid w$$

$$w \rightarrow bw \mid \epsilon$$

$$D = \{a^i b^j c^k / i, j, k \geq 0 \text{ and } i+j=k\}$$

$$S \rightarrow aSc \mid x$$

$$x \rightarrow bxc \mid \epsilon$$

$$E = \{w \in \{[, \cdot, \cdot]\}^* / w \text{ has properly balanced brackets}\}$$

$$S \rightarrow sS \mid [s] \mid \epsilon$$

(5/6)

* Pushdown Automata (PDA)

* Σ^+ is used to represent an irregular language equivalent to a CFG.

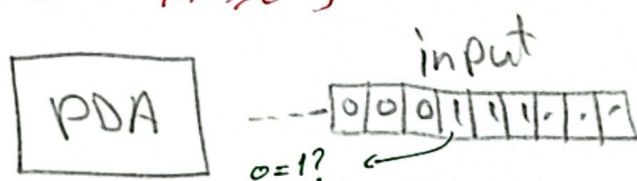
* PDA = FA + stacks (built-in)

* A language is a CFL i.e.

1. There is a CFG that generates this language

2. There is a PDA that recognizes this language

ex: $L = \{0^n 1^n \mid n \geq 0\}$



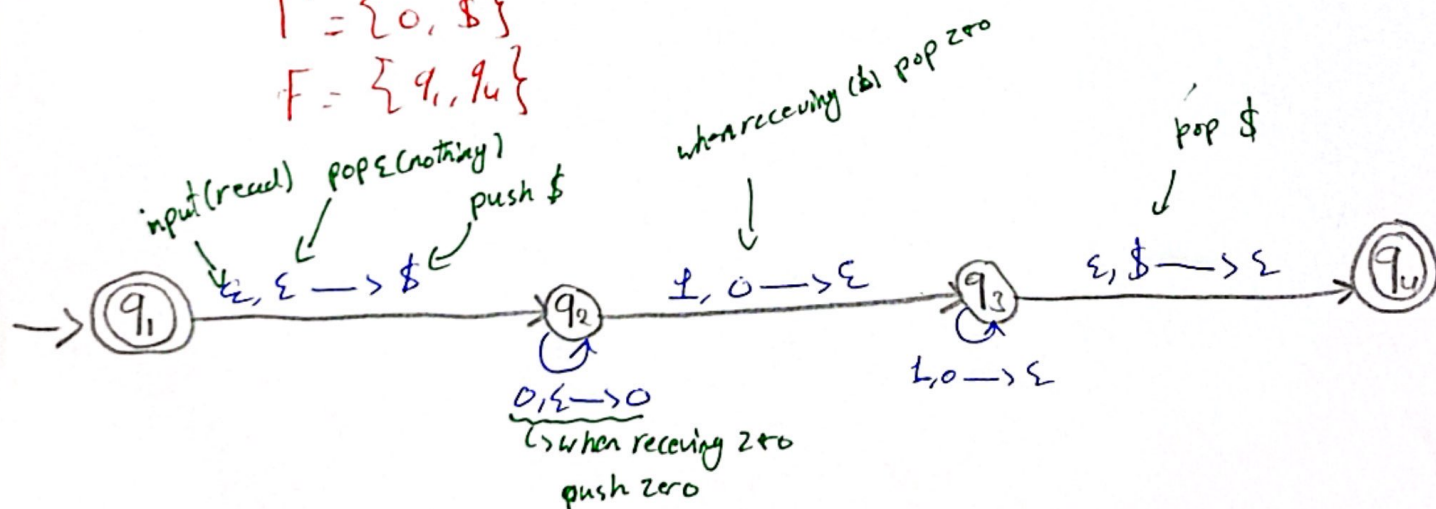
push zeros Then when receiving ones pop zero and compare

ex: $Q = \{q_1, q_2, q_3, q_4\}$

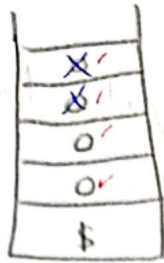
$\Sigma = \{0, 1\}$

$\Gamma = \{0, \$\}$

$F = \{q_1, q_4\}$



ex: 0000xx



* Formal Definition:

$PDA = 6\text{-tuple } (Q, \Sigma, \Gamma, \delta, q_0, F)$

Q : start state

Σ : input Alphabet

Γ : stack Alphabet

To be continued